Human-AI Collaboration in Scientific Writing and Research Workflows

Abstract

The integration of artificial intelligence into the scientific discovery process is undergoing a profound paradigm shift, transitioning from the use of isolated computational oracles to the deployment of autonomous, agentic workflows. This survey provides a comprehensive examination of human-AI collaboration across the entire scientific lifecycle, encompassing literature review, hypothesis generation, autonomous experimentation, academic writing, and peer review validation. Recent advancements in Large Language Models (LLMs) and multi-agent systems have enabled frameworks capable of executing end-to-end research pipelines, fundamentally altering traditional conceptions of intellectual labor and academic authorship. However, the deployment of these systems reveals a complex landscape of capabilities and critical limitations. While AI agents demonstrate profound proficiency in structured data synthesis, code generation, and protocol optimization, empirical evidence suggests they currently narrow scientific exploration by favoring the recombination of existing methods over the generation of fundamentally novel research paradigms. Furthermore, the proliferation of AI-assisted scientific writing has exacerbated the risk of citation hallucinations and epistemic debt, necessitating robust verification frameworks and a recalibration of how evidence-licensed claims are generated and validated. By synthesizing contemporary architectures, empirical benchmarks, and sociotechnical governance models, this survey delineates the trajectory of human-AI co-creation in research workflows and underscores the enduring necessity of human-in-the-loop oversight to ensure the integrity, diversity, and reliability of scientific knowledge production.

Introduction

Historically, scientific discovery and academic writing have been conceptualized as exclusively human-driven cognitive exercises. The traditional scientific method relies on the researcher's ability to synthesize prior literature, intuitively identify knowledge gaps, formulate testable hypotheses, execute controlled experiments, and rigorously communicate the findings[1]. In recent decades, artificial intelligence has served primarily as a computational tool—a static oracle deployed for highly specific tasks such as protein folding prediction, statistical regression, or basic grammatical correction[3]. However, the advent of sophisticated foundation models and agentic architectures has catalyzed a qualitative leap from "AI for Science" to "AI as Scientist"[4].

The evolution of AI autonomy in research workflows can be classified into three distinct tiers of operational capability. At Level 1 (Tool), AI operates under direct human supervision, executing discrete, well-defined tasks within a single stage of the scientific method, such as literature summarization or generating code snippets for data processing. At this level, the primary goal is to enhance researcher efficiency, with outputs strictly requiring human validation. At Level 2

(Analyst), AI systems function as passive agents capable of independent information processing, data modeling, and analytical reasoning. While operating within human-defined boundaries, Level 2 systems manage sequences of tasks, such as identifying trends in experimental datasets or refining computational models. At Level 3 (Scientist), AI systems achieve high autonomy, operating as active agents capable of orchestrating multiple stages of the research cycle—from hypothesis formulation to experimental execution and manuscript generation—with minimal human intervention[6].

This transition has profound implications for the nature of scientific work. Large Language Models are no longer merely transcribing human thought; they are actively participating in the epistemic process, altering the dynamics of cognitive engagement, and shifting the role of the human researcher from primary creator to supervisor, editor, and validator[1]. The distinction between "AI-written" and "human-written" text is becoming less about the origin of the content and more about how humans and machines collaborate to optimize the complex cognitive processes of writing[1]. As these autonomous systems become increasingly capable of generating coherent, substantive research manuscripts for minimal computational costs, the scientific community is confronted with unprecedented opportunities for productivity acceleration[9].

However, this productivity frontier exposes severe vulnerabilities regarding academic integrity, the homogenization of ideas, and systemic hallucinations[9]. AI systems, despite their fluency, frequently fabricate results, miss hidden methodological errors, and fail to judge novelty reliably when detached from physical grounding[12]. Consequently, establishing a unified understanding of how humans and AI actually collaborate across the scientific discovery life-cycle is paramount[13].

This survey systematically deconstructs the human-AI collaborative research workflow into coherent architectural and functional domains. It examines the integration of AI in literature review, hypothesis generation, autonomous experimentation via Self-Driving Laboratories, collaborative writing, and peer review simulation. Furthermore, it addresses the critical challenges threatening the reliability of AI-augmented science, exploring the necessity for rigorous citation verification frameworks and the adoption of a formal calibration process to ensure that AI-generated scientific claims are explicitly licensed by underlying evidence.

Methods

The integration of AI into scientific workflows can be structurally mapped to the chronological stages of the scientific method. To provide a systematic synthesis, this section organizes the research area into distinct method families, analyzing the algorithmic architectures, multi-agent frameworks, and human-AI interaction models governing each stage of the discovery pipeline.

AI-Assisted Literature Review and Knowledge Extraction

Scientific discovery is an iterative process that relies heavily on the systematic exploration and synthesis of prior work to identify methodological trends and gaps in existing knowledge[14]. The exponential growth of scientific publications has rendered traditional manual literature reviews increasingly intractable, prompting the development of AI-assisted discovery pipelines[14].

Modern AI tools intervene at multiple phases of the literature review process, including semantic discovery, predictive screening, structured synthesis, and ongoing monitoring[16]. Traditional retrieval systems relied largely on sparse lexical matching (e.g., BM25), which often failed to capture conceptual relationships across disparate terminologies. Contemporary AI-assisted literature frameworks utilize hybrid retrieval strategies, combining dense semantic encoders with cross-encoder re-ranking to map conceptual relationships between distinct research fronts[17]. To process massive scientific corpora, these pipelines rely on layout-aware parsing techniques that reconstruct logical document structures—such as sections, figures, tables, and citation graphs—yielding indexed repositories that preserve both textual content and structural semantics essential for downstream reasoning[17].

A prominent architecture driving this domain is Agentic Retrieval-Augmented Generation (RAG). Systems such as PaperQA exemplify this approach, operating as retrieval-augmented generative agents specifically tailored for scientific research[19]. Unlike basic search tools that merely return a list of documents, Agentic RAG frameworks actively synthesize multi-document insights. If an initial retrieval yields insufficient data to answer a complex scientific query, the agent autonomously rephrases the query, loops back to the retrieval phase, gathers additional evidence, and formulates a synthesized response complete with inline citations[17].

To further enhance domain-specific comprehension, frameworks such as SciLitLLM employ continual pre-training and supervised fine-tuning strategies to infuse specialized scientific knowledge into LLMs, improving their instruction-following capabilities for complex document classification and summarization tasks[14]. Similarly, evaluation benchmarks like LitSearch and ResearchArena have been developed to systematically assess the performance of LLM-based agents in navigating complex literature search queries and structuring academic surveys[14].

Literature Review Stage	**AI Intervention Mechanism**	**Key Technologies / Frameworks**
Discovery	Semantic search and automatic query expansion	Dense semantic encoders, Vector embeddings, SciLitLLM
Screening	Predictive relevance classification	Natural Language Processing (NLP) classifiers, Active learning
Synthesis	Structured data extraction and thematic clustering	Agentic RAG, PaperQA, Large Language Models (LLMs)
Monitoring	Automated alerting on new	Knowledge graphs,

	relevant publications	Profile-based database watchers

Despite these advanced capabilities, relying on fully autonomous synthesis presents substantial risks. Empirical observations indicate that AI models frequently overlook methodological weaknesses shared across multiple studies, fail to distinguish between reported benchmark performance and real-world applicability, and exhibit a pervasive tendency to smooth over critical contradictions in the literature[20]. Consequently, the most robust literature review pipelines mandate a human-in-the-loop (HITL) architecture. In these collaborative workflows, AI agents are utilized for the mechanical stages—such as structured extraction, initial thematic clustering, and abstract screening—while the human researcher maintains responsibility for critical interpretation, contradiction mapping, and the manual validation of every generated citation[20].

Hypothesis Generation and Problem Formulation

The formulation of novel, testable hypotheses represents the creative core of scientific discovery[2]. AI frameworks are increasingly deployed to automate this process by uncovering latent connections within heterogeneous scientific corpora. This phase typically involves conceptual fusion and trend extrapolation, where dense retrieval, embedding-based clustering, and knowledge graph reasoning are utilized to reveal thematic patterns and analogical transfers across disparate knowledge domains[4].

Advanced hypothesis generation systems operate through sophisticated multi-agent collaboration, moving beyond zero-shot generation to structured brainstorming and cross-domain recombination. For instance, the SciAgents framework utilizes a tripartite, knowledge-graph-grounded architecture comprising an Assistant Agent that drafts contextual subgraphs, an Ontologist Agent that defines nodes and interprets edges, and Scientist Agents that iteratively propose, elaborate, and critique detailed hypotheses[5]. Similarly, the ResearchAgent system automates problem definition, method proposition, and experimental design by leveraging an academic graph and a conceptual knowledge store. It explicitly mimics the human peer review process by utilizing multiple LLM-based reviewing agents that provide iterative feedback based on evaluation criteria elicited from actual human judgments[24].

While the computational throughput of these systems is extraordinary, recent large-scale empirical evaluations raise critical epistemological questions regarding the true novelty of AI-generated hypotheses. A comprehensive study evaluating 37,802 scientific ideas generated by four AI research-agent frameworks across machine learning domains revealed a pronounced "narrowing" effect on scientific exploration[26]. The analysis yielded four consistent and concerning patterns:

1. AI-generated ideas exhibit significantly higher concentration and lower diversity than human-authored papers within the same research domains.
2. AI-generated concepts remain much closer to their seed literature than subsequent

human-led follow-on research.

3. Human-authored papers exhibiting high semantic similarity to AI-generated ideas tend to receive lower subsequent citation impact, suggesting lower scientific utility.
4. When AI-generated ideas do diverge from prior work, they overwhelmingly do so by recombining existing technical methods rather than formulating fundamentally novel research questions or paradigms[26].

These findings indicate that while LLMs excel at local elaboration and conceptual interpolation, they currently lack the capacity for genuine epistemological extrapolation. The propensity of AI systems to optimize toward annotator consensus during Reinforcement Learning from Human Feedback (RLHF) further homogenizes generated ideas, discouraging unconventional scientific leaps[29]. To counteract these limitations, researchers propose leakage-aware evaluation architectures, such as the DP-Scientist framework. This approach utilizes cutoff-based protocols like RetroSci-Explore to strictly separate visible generation-time records from hidden future records, ensuring a more rigorous assessment of an AI agent's actual predictive novelty and mitigating the illusion of discovery caused by data contamination[30].

Experimental Design and Autonomous Execution (Self-Driving Laboratories)

The transition from a theoretical hypothesis to empirical validation requires the translation of high-level scientific goals into concrete, executable protocols[2]. In physical sciences such as chemistry, materials science, and biology, this is increasingly facilitated by Self-Driving Laboratories (SDLs). SDLs represent closed-loop platforms capable of designing, executing, and analyzing experiments with minimal human input, merging robotic automation with AI-driven decision-making to accelerate knowledge generation[32].

Autonomous experimentation frameworks generally organize the capabilities required for agent-enabled SDLs across a five-module architecture: Comprehension, Design, Execution, Analysis, and Optimization[34]. To navigate complex, high-dimensional design spaces, SDLs rely heavily on Bayesian optimization, active learning, and reinforcement learning. These techniques allow the system to sample experimental parameters efficiently, maximizing information gain while minimizing resource expenditure in environments characterized by noisy feedback and strict safety constraints[35].

Recent advancements have witnessed a shift from isolated automation to the integration of LLMs as central, agentic orchestrators within these physical laboratories. Prominent milestones in this transition include:

- **Coscientist:** An artificial intelligence system driven by GPT-4 that autonomously designs, plans, and performs complex chemical experiments. By integrating large language models with external tools—such as internet and documentation search, Python code execution, and cloud laboratory APIs—Coscientist has successfully automated the reaction optimization of palladium-catalyzed cross-couplings[37].
- **ChemCrow:** An LLM chemistry agent augmented with 18 expert-designed tools. Recognizing that standard LLMs struggle with domain-specific chemical reasoning (e.g., predicting reaction outcomes or converting IUPAC names), ChemCrow offloads these

tasks to specialized computational modules. This agentic system autonomously planned and executed the syntheses of insect repellents and organocatalysts on a robotic platform, bridging the gap between natural language reasoning and physical execution[39].

- **ChemToolAgent (CTA):** Building upon the ChemCrow architecture, CTA integrates an expanded suite of 29 specialized tools within a ReAct framework. By categorizing tools into general reasoning, molecular analysis, and reaction prediction, CTA demonstrates how expansive tool-use can enhance AI efficacy across a broader spectrum of complex chemical scenarios, including comprehensive database querying via PubChem integration[42].

SDL / Agent System	Primary Domain	Core Modality / Tool Integration	Key Scientific Achievements
Coscientist	Chemistry (Reaction Optimization)	GPT-4, Web Search, Code Execution, Cloud Lab API	Autonomous optimization of palladium-catalyzed cross-couplings.
ChemCrow	Synthesis Planning & Drug Discovery	LLM augmented with 18 specialized computational chemistry tools	Autonomous synthesis of insect repellents and multiple organocatalysts.
ChemToolAgent	General Chemistry Tasks	ReAct framework utilizing 29 categorized molecular and reaction tools	Enhanced performance on molecular property prediction and complex semantic queries.

Despite these high-profile successes, the fully autonomous laboratory remains an aspirational benchmark. Current SDLs face significant bottlenecks in physical execution and contact-rich manipulation. To systematically address these limitations, researchers have introduced the ADePT framework, which evaluates robotic capabilities across four core dimensions: Adaptability and learning, Dexterity, Perception, and Task complexity[33]. Furthermore, empirical evaluations of tool-augmented agents reveal that excessive tool access can sometimes introduce additional cognitive complexity, occasionally hindering the LLM's logical consistency[42]. Consequently, contemporary best practices advocate for a synergistic human-AI-SDL collaborative model. In this paradigm, robotic agents manage high-throughput execution and structured data handling, while human researchers maintain exclusive responsibility for scientific framing, objective translation, interpretation of multi-objective trade-offs, and critical safety supervision[34].

Scientific Writing and Human-AI Co-Creation

The formalization of scientific discoveries into academic manuscripts is undergoing a radical transformation. Writing is no longer viewed solely as a transcription of human thought; it is an interactive process encompassing profound cognitive, social, and affective dimensions[1]. As generative AI tools become embedded in word processors, the cognitive architecture of writing is fundamentally altered. Modern writing processes are typically categorized into four cognitive stages: planning, translating (drafting), reviewing, and monitoring[45].

Research into human-AI collaborative writing reveals that writers' desired levels of AI intervention vary significantly across these cognitive stages. Content-centric writers, such as academic researchers, fiercely prioritize human ownership during the planning and ideation phases, viewing these as the locus of intellectual contribution. However, they frequently delegate translational and reviewing tasks to AI to alleviate cognitive load. Conversely, form-centric creative writers often prefer maintaining strict control over the stylistic translation of the text[45].

The most extreme manifestation of AI in scientific writing is the pursuit of end-to-end autonomous manuscript generation. "The AI Scientist" represents a comprehensive framework that seamlessly executes ideation, literature search, experiment planning, code modification, figure generation, and manuscript drafting. Operating primarily within machine learning subfields—such as diffusion modeling, transformer-based language modeling, and learning dynamics—this system utilizes the open-source coding assistant Aider to implement ideas into full, LaTeX-formatted papers for a computational cost of approximately $15 per manuscript[9]. The AI Scientist utilizes concepts from evolutionary computing, treating ideas as mutable entities that are iteratively self-assessed for novelty and feasibility before execution[10].

While systems like The AI Scientist demonstrate remarkable automation, they highlight severe qualitative limitations. AI-generated texts consistently exhibit stylistic homogenization, diminish cultural nuances, and often lack the deep, tacit domain knowledge required to fully interpret the causal mechanisms behind an algorithm's success or failure[10]. In mathematical research, for example, detailed case studies using AI assistants to discover novel error representations for Hermite quadrature rules revealed that while AI excels at algebraic manipulation and systematic proof exploration, every step required rigorous human verification to prevent the generation of "plausible nonsense"[21].

Furthermore, current AI writing tools are predominantly designed for isolated, single-user interactions, creating severe friction in collaborative academic environments where co-authors must leave shared workspaces to interact with LLMs[51]. To address this, the Human-Computer Interaction (HCI) and Computer-Supported Cooperative Work (CSCW) communities advocate for "Interaction-Centered Intelligence." This paradigm proposes integrating compositional structures (spatial, temporal, and narrative) directly into collaborative writing environments, ensuring that human agency and ownership are preserved without succumbing to the excessive anthropomorphism of the AI agent[8].

Similarly, in qualitative research methodologies, systems like ChatQDA and QualAnalyzer have been developed to support human-AI co-creation during the labor-intensive open coding process[54]. These tools enforce "atomistic LLM analysis," processing data segments independently and preserving the prompts and outputs for every unit to maintain reflexive accountability, reflecting a "conditional trust" where users value AI for surface-level extraction but question its interpretive nuance[54].

AI in Peer Review and Validation

The academic peer review system, historically the primary mechanism for scientific quality control, is currently straining under exponential submission growth and severe reviewer fatigue[56]. In response, the academic community is rapidly exploring Large Language Models as tools for both automated peer review simulation and active evaluation assistance[58].
Several advanced multi-agent frameworks have been developed to deconstruct and simulate the intricate dynamics of the peer review pipeline:

- **AgentReview:** This framework models the peer review process across a structured five-phase pipeline: Reviewer Assessment, Author-Reviewer Discussion, Reviewer-AC (Area Chair) Discussion, Meta-Review Compilation, and Paper Decision. By simulating tens of thousands of peer review documents, AgentReview enables controlled experiments on latent social variables. The simulations reveal how phenomena like "Groupthink" (where biased reviewers drive a panel to consensus, causing a measurable spillover decrease in ratings) and "Altruism Fatigue" (where the presence of an irresponsible reviewer causes a sharp drop in overall panel engagement) manifest in automated academic evaluations[59].
- **DIAGPaper:** Addressing the limitations of generic, single-agent AI reviews, DIAGPaper is a multi-agent framework utilizing three tightly integrated modules. A *Customizer* module assigns criterion-specific expertise to different reviewer agents based on the paper's content. A *Rebuttal* module introduces author agents that engage in structured debate with reviewer agents to challenge and filter out invalid critiques. Finally, a *Prioritizer* module ranks the severity of validated weaknesses, ensuring that feedback is specific, actionable, and user-oriented rather than merely superficially plausible[61].
- **PeerCheck:** This system systematically investigates the divergence between human and AI review criteria. Empirical analyses reveal that while human reviewers heavily prioritize methodological rigor and experimental execution, LLMs consistently emphasize theoretical aspects. PeerCheck utilizes Chain-of-Thought (CoT) prompting to align LLM outputs closer to human evaluation standards. Notably, the framework uncovered a "RAG paradox," wherein injecting external retrieval data occasionally degraded the coherence and quality of the LLM's evaluative reasoning, suggesting that excessive context can disrupt the model's analytical focus[57].

System	Architecture	Core Function / Contribution	Key Finding / Phenomenon Addressed

AgentReview	5-phase multi-agent simulation (Reviewers, Authors, ACs)	Disentangling latent social factors in peer review	Quantifies "Groupthink" and "Altruism Fatigue" in review panels.
DIAGPaper	Multi-agent (Customizer, Rebuttal, Prioritizer modules)	Diagnosing valid, specific paper weaknesses	Author-agent debate filters out invalid, superficially plausible critiques.
PeerCheck	LLM vs Human comparative framework	Aligning AI reviews with human criteria via CoT	Identifies the "RAG paradox" where external data degrades review quality.

While LLMs demonstrate utility in automating objective tasks—such as reproducibility verification, reference formatting checks, and policy compliance—relying on them for comprehensive scientific judgment introduces profound systemic risks[56]. If uniform LLM architectures are widely adopted across a discipline, the peer review process risks devolving into a procedural formality, reinforcing algorithmic biases, prioritizing generic criticism patterns, and failing to recognize nuanced, paradigm-shifting methodological innovations[56].

Challenges and Prospects

The rapid integration of generative AI into scientific workflows presents a host of epistemological, ethical, and practical challenges that threaten the structural integrity of the scholarly record. Addressing these issues requires both rigorous technical interventions and a fundamental shift in scientific governance models.

Epistemic Integrity and Citation Hallucinations

The most immediate and pervasive threat posed by LLM-assisted scientific writing is the generation of hallucinated citations. These are references that appear stylistically and contextually plausible but either contain corrupted metadata (e.g., mismatched authors and DOIs) or correspond to completely fabricated papers[64]. A recent, massive-scale computational audit of 111 million references across 2.5 million papers in preprint servers and open-access repositories (including arXiv, bioRxiv, and PubMed Central) estimated that over 146,000 hallucinated citations were introduced into the scientific literature in 2025 alone[11].

The consequences of these fabrications are profound. They disproportionately assign credit to already prominent and male scholars, thereby reinforcing existing demographic inequities in scientific recognition[11]. Furthermore, because mistakes in highly-cited works replicate across the literature through a process of "citation mutation," these fabrications threaten to permanently pollute the knowledge network, wasting resources and undermining the self-correcting nature of scientific inquiry[11].

To combat this epidemic, the research community is rapidly developing specialized retrieval-grounded detection frameworks:

- **CiteCheck:** This hybrid framework treats citation hallucination detection as a three-class severity problem. It retrieves candidate publications from external scholarly databases and uses a structured LLM verifier to classify citations as *Exact matches*, *Minor hallucinations* (where the paper exists but metadata is corrupted), or *Major hallucinations* (entirely fabricated entities pointing to no matching scholarly work)[64].
- **DeepSciVerify:** Focusing specifically on scientific claim–citation alignment, this system employs a two-stage escalation pipeline. It first attempts to verify a claim using only the cited paper's abstract (enabling an efficient early exit). If the evidence is inconclusive, it escalates to passage-level retrieval and full-text semantic search. This selective escalation leverages complementary behaviors across LLMs, optimizing both computational efficiency and verification accuracy[70].

Institutional responses have also escalated rapidly. Major preprint repositories, including arXiv, have instituted stringent policies, including mandatory one-year submission bans for authors found submitting manuscripts with unverified, LLM-generated fabricated references, reflecting the severe threat these hallucinations pose to scholarly infrastructure[66].

The Calibration Turn and Evidence-Licensed Claims

Beyond explicit hallucinations and fabricated references, a deeper epistemological issue is the misalignment between the authoritative language used by AI agents and the actual empirical evidence supporting their outputs. The field is increasingly advocating for a "Calibration Turn" in AI-assisted research. This paradigm argues that generating a hypothesis and running a simulation is insufficient for knowledge creation; the final output must be rigorously calibrated to ensure that the scientific claims made are strictly licensed by the underlying evidence[73].

When an AI system generates a claim that exceeds the evidentiary support provided by the data, it creates an "epistemic debt" and a measurable claim-evidence gap[75]. Future autonomous discovery systems must incorporate a explicit "claim calibration operator." This mechanism maps raw AI claims, experimental evidence, and validation data to a maximal licensed frontier, automatically downgrading the certainty of the system's language when evidence is scarce or noisy[75].

Furthermore, to maintain methodological accountability, researchers propose adopting the paradigm of treating "AI as a Research Object" (AI-RO). Under this view, the legitimacy of an AI-assisted paper depends on how the model's use is documented and made accountable. AI interactions are treated as structured, inspectable components of the research process, requiring explicit documentation of system constraints, audit checklists, and verifiable provenance records (such as the RO-Crate structure) to link claims directly to sources, rather than becoming entangled in philosophical debates about whether AI constitutes a co-author[77].

Sociotechnical Governance and Institutional Readiness

The transition to agentic scientific discovery cannot be managed by algorithmic safeguards alone; it necessitates robust sociotechnical frameworks and institutional redesign. Theoretical frameworks such as Pingquanqi advocate for interaction design protocols embedded directly at the agent orchestration layer. These protocols address the cognitive cost asymmetry between human supervisors and autonomous agents, enforcing "economic alignment" to ensure agents do not maximize user attention at the expense of rigorous scientific contemplation[79].

Additionally, autonomous agents suffer from inherent biases in their training data, which overwhelmingly encodes successful, published literature while systematically discarding the tacit knowledge, procedural friction, and unpublished dead-ends of daily laboratory practice[29]. To correct this, the community must build open laboratory datasets that capture failure knowledge. Furthermore, to prevent the proliferation of false positives generated by high-throughput AI systems, researchers strongly advocate for the centralized public preregistration of all AI-generated hypotheses and validation experiments. Modeled on clinical trial protocols and tracked via persistent identifier schemes (e.g., AICID), this infrastructure would enable the community to audit the evolutionary trajectory of AI-generated theories and hold hybrid human-machine knowledge chains accountable[29].

Conclusion

The advent of human-AI collaboration in scientific workflows represents a profound inflection point in the history of academic knowledge production. Frameworks such as Agentic RAG for literature synthesis, Self-Driving Laboratories for physical experimentation, and multi-agent peer review simulators demonstrate that AI has definitively evolved from a passive computational tool into an active epistemic partner. Systems like The AI Scientist prove that end-to-end automated research is mechanically feasible and economically highly scalable. However, this survey reveals that profound systemic vulnerabilities accompany this transition toward automation. While AI agents drastically increase research throughput, empirical evidence indicates they exhibit a pervasive tendency to narrow scientific exploration, heavily favoring conceptual interpolation and methodological recombination over genuine paradigm-shifting innovation. Furthermore, the proliferation of hallucinated citations and the accumulation of epistemic debt threaten the foundational reliability of the scholarly record. Ultimately, realizing the full potential of AI-assisted scientific discovery requires moving beyond the blind pursuit of total autonomy. The future of robust scientific inquiry lies in the "Scholarly HI-AI Loop"—a meticulously calibrated, hybrid workflow where automated systems handle massive data synthesis, high-throughput physical experimentation, and structural formatting, while human researchers serve as the irreplaceable epistemic anchors. By enforcing rigorous citation verification, demanding evidence-licensed claims, and maintaining ultimate interpretative authority, the scientific community can harness the unprecedented capabilities of agentic AI without compromising the foundational integrity of scientific truth.

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